

What is an Embedding?

An **embedding** is a way of converting data (such as words, sentences, images, or users) into a **dense numerical vector** that captures its meaning and relationships.

Instead of representing a word as a sparse vector like One-Hot Encoding:

Word	One-Hot Vector
Cat	[1,0,0,0]
Dog	[0,1,0,0]
Car	[0,0,1,0]
Bus	[0,0,0,1]

we represent it as:

Word	Embedding Vector
Cat	[0.82, -0.15, 0.67]
Dog	[0.79, -0.12, 0.70]
Car	[-0.55, 0.91, -0.22]
Bus	[-0.60, 0.88, -0.18]

These numbers are **learned automatically** during training.

